PARALLEL MINDS: A THEMATIC LITERATURE REVIEW OF ARTIFICAL INTELLIGENCE IN EDUCATION

By

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Abstract

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This thesis explores the complex interplay between artificial intelligence (AI) and human cognition, focusing on AI's impact on motivation, identity, and cognitive load in educational contexts. Through a detailed examination of key psychological frameworks, including Self-Determination Theory (SDT), Cognitive Load Theory (CLT), and Identity Theory, the study highlights both the benefits and risks of AI integration in learning environments. AI systems offer immense potential to enhance learning by fostering student engagement, reducing cognitive load, and providing personalized feedback. However, they also pose significant risks, such as diminished autonomy, the homogenization of individual experiences, and over-reliance on AI, which can undermine intrinsic motivation and weaken identity formation.

A thematic literature review, guided by a modified PRISMA protocol, examined recent empirical studies on AI's role in education, revealing the dual nature of AI's influence: while AI can alleviate cognitive strain and support personalized learning pathways, it can also overwhelm students with information and foster dependency on technology for problem-solving. The findings underscore the importance of designing human-centered AI systems that complement, rather than replace, human cognition. This thesis advocates for a balanced approach to AI development, calling for deeper investments in cognitive science to ensure that AI enhances creativity, autonomy, and intellectual growth without compromising the richness of human experience. The study concludes with recommendations for AI literacy education, emphasizing the need for critical engagement with AI technologies to maintain personal agency and cognitive independence in an increasingly AI-mediated world.

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To my parents, Dirk & Shelly Fanning, whose constant encouragement and belief in my academic journey have inspired me to reach beyond what I thought was possible.

CHAPTER ONE: AI IN OUR IMAGE: THE IRONY OF OUTPACED UNDERSTANDING

Alex, a project manager at a bustling tech company, relies on AI-driven tools to organize their schedule, filter emails, and even generate meeting summaries. Every morning, they check a dashboard that forecasts their productivity and suggests optimal times for focused work. For Alex, AI has become an invisible assistant, subtly shaping their day without much conscious thought.

The rapid integration of artificial intelligence (AI) into contemporary society has profoundly transformed daily life, influencing our interactions, work, and thought processes. This seamless integration underscores the reach of AI, as it effortlessly melds with routines in ways previous technologies never could (Harvard School of Engineering and Applied Sciences, 2021). AI technologies, ranging from voice-activated personal assistants to complex algorithms driving autonomous vehicles and analyzing vast datasets, have seamlessly woven into our routines (Johnson, 2020). This swift adoption contrasts with earlier technological advancements, such as television, the internet, and cell phones, which faced slower integration due to higher initial costs and accessibility barriers. For instance, cell phone usage in U.S. households increased from 10% in 1994 to 96% in 2019, highlighting a gradual adoption curve compared to the rapid proliferation of AI technologies (Digital Adoption, 2019). AI forges a different path than previous advancements as the barriers to access are significantly lower due to the widespread integration of the internet and the prevalence of connected devices in nearly every aspect of modern life. Smartphones, tablets, and computers have become staples in modern society, making access to AI-driven tools almost effortless. This ease of access accelerated AI's impact as more people can seamlessly incorporate it into their routine without significant financial or technical hurdles. These advancements have generated excitement and unease, as AI continues to emulate cognitive tasks that were once thought uniquely human. AI models have started to break into spheres of content creation, creativity, artistic expression, and conversation, all of which go beyond what was originally hypothesized to be possible. This intersection between technological progress and cognitive replication reveals a paradox: while AI advances with astonishing speed, our understanding of the underlying cognitive processes and mechanisms remains incomplete.

At this juncture, we encounter a profound irony. AI's capacity to replicate certain aspects of human cognition—such as decision-making, learning, and even creative processes—exposes a fundamental gap in our comprehension of the mind itself. Alan Turing, one of the pioneers of AI, famously suggested that a machine could simulate any conceivable process, raising questions about the nature of intelligence and the boundaries of computation (Turing, 1950). However, as AI advances, it prompts us to reflect on whether our efforts to create intelligent machines are merely outpacing our understanding of the human cognition they imitate. In essence, the capabilities of AI highlight the limitations of our own knowledge, as we build systems that operate in ways we may not fully grasp (Crevier, 1993).

In creating AI that mirrors human traits, we are compelled to confront questions about the nature of identity. What does it mean to replicate human attributes in a machine, and how does this affect our sense of self? As AI begins to influence not only our cognitive processes but also our perceptions of who we are, it blurs the line between human and machine, challenging our assumptions about autonomy, agency, and the essence of personhood (Brynjolfsson & McAfee, 2017). The development of AI has thus become a double-edged sword: it enables us to understand certain aspects of cognition while simultaneously revealing the limits of that understanding. This paradox forces us to question whether we are truly gaining insight into human cognition or simply complicating our perception of it.

The Intersection of Identity, Motivation, Cognitive Load, and AI

As Alex unwinds in the evening, they often scroll through their social media feeds, unaware of how AI algorithms curate content that aligns with their browsing history and previous interactions. This AI-mediated experience creates a bubble of familiarity, reinforcing Alex's existing perspectives and interests. With each click, the AI refines its predictions, subtly guiding Alex's views and online interactions, raising questions about how much autonomy they truly have in this digital landscape. For Alex, the experience is both comfortable and confining, a reminder of AI's silent hand in shaping identity.

This tension between innovation and comprehension is especially evident when we consider AI's impact on human identity. Identity Theory provides a framework for understanding how individuals form and maintain their sense of self—a process inherently linked to social roles and personal experiences (Burke & Stets, 2009). As AI systems increasingly mediate these experiences, they influence how we perceive ourselves and relate to others. This effect is especially pronounced in digital spaces where AI algorithms determine the content we see, shaping our beliefs and reinforcing our identities. Alex's experience is commonplace in the modern social media space, as many social media platforms employ AI-driven recommendation systems that curate content aligned with our preferences, potentially leading to echo chambers that reinforce specific aspects of identity while limiting exposure to diverse perspectives (Stryker & Burke, 2000). Beyond content curation, social media platforms are populated by bots – automated accounts programmed to perform tasks such as liking, sharing, and commenting on posts across various platforms (ICUC Social, n.d.). These bots are often programmed to mimic human behavior, interacting with users and content in ways that feed data into algorithms and shape the media we consume. By amplifying specific narratives and trends, bots can subtly influence public opinion and shape our perceptions, further reinforcing echo chambers that align with particular beliefs (ICUC Social, n.d.). This cycle of algorithmic content curation and bot-driven interactions creates a feedback loop that affects how individuals form and maintain their identities within digital spaces. As a result, AI's presence in these environments not only reflects our existing identity, but actively participates in molding them, raising questions about authenticity, agency, and the extent to which we control our own self-concept in an AI-mediated world.

The rise of AI necessitates a deeper exploration of motivation and cognitive load—two fundamental components that drive learning, productivity, and personal growth. Motivation, described by Ryan and Deci (1985) as the force that energizes and directs behavior toward goals, is fundamental to both educational and professional success. Building on this foundational understanding, their Self-Determination Theory (SDT) posits that motivation exists on a continuum, ranging from intrinsic motivation—rooted in personal satisfaction and interest—to extrinsic motivation, driven by external rewards or pressures. In educational settings, motivation is critical for fostering student engagement, resilience, and long-term achievement. AI intersects with motivation in complex ways. For some, AI-driven tools may be inherently motivating due to their interactivity and personalized nature, echoing the intrinsic motivation found in social interactions (Zhang & Aslan, 2021; García-Martínez et al., 2023). Others argue that AI also allows us to explore the nuances of human motivation by mirroring our own behaviors and preferences back to us, prompting reflection on what truly drives us as people (Zhang & Liu, 2022). In this sense, AI does not just facilitate motivation, but also has the potential to challenge and broaden our understanding of it. Unlike humans, AI lacks intrinsic motivation and never tires, constantly reflecting our goals and actions which may provoke us to question and refine our own motivations in response (Ryan & Deci, 2000). Without motivation, even the most advanced AI tools and technologies may fail to yield meaningful outcomes, as students may not possess the necessary drive to engage with and benefit from these resources (Talan, 2021).

Motivation is crucial not only in educational contexts but also in how individuals navigate challenges and derive meaning from their work, hobbies, and personal pursuits. In professional settings, motivated employees are more likely to display creativity, initiative, and resilience (Runan & Deci, 2000). Yet, as AI increasingly automates routine tasks and provides decision support, it reshapes these motivational dynamics by potentially reducing the intrinsic satisfaction derived from manual or cognitive labor (Brynjolfsson & McAfee, 2017). While AI can enhance efficiency and free individuals to focus on more complex, fulfilling tasks, it may also inadvertently reduce engagement by removing elements of challenge and personal involvement.

Cognitive Load Theory developed by Sweller (1988) asserts that human cognitive resources are limited, and that overloading this capacity can impede learning and problem-solving. This concept becomes particularly relevant in AI-mediated environment as AI systems handle an increasing portion of cognitive tasks, they alter the demands placed on human cognition. CLT categorizes cognitive load into three types: intrinsic, extraneous, and germane. Intrinsic load relates to the inherent complexity of the task, extraneous load is associated with how information is presented, and germane load pertains to the mental effort required to process and internalize information. Effective instructional design seeks to manage these types of load to maximize learning outcomes, a principle increasingly relevant in AI-driven educational environments (Sweller, Ayres, & Kalyuga, 2011). AI Technologies can adaptively adjust the presentation of information to minimize extraneous load by customizing learning pathways based on individual needs and preferences.

This interplay between motivation and the nature of tasks performed highlights the importance of Cognitive Load Theory (CLT) in understanding how AI affects our cognitive resources. In practical terms, AI can be a powerful tool for managing cognitive load, particularly in settings where complex information must be processed efficiently. Adaptive learning systems, for example, can adjust the difficulty of tasks based on a learner's performance, maintaining an optimal cognitive load that promotes engagement without causing overload (Mayer & Moreno, 2003). This capacity to tailor content to individual needs has significant implications for education and professional development, as AI can support learners in focusing on essential information while minimizing distractions. However, if not carefully designed, AI systems can also contribute to cognitive overload by overwhelming users with information or by automating tasks to the point where individuals become cognitively dependent on these technologies (Ayres & Paas, 2009).

Beyond the Classroom

Beyond educational contexts, the implications of motivation and cognitive load extend into all areas of life. In personal development, motivation drives individuals to set goals, engage in learning activities, and persist in the face of setbacks. It is a critical component of resilience, helping people navigate challenges and pursue personal growth. Cognitive load affects how individuals process and retain information in a world where they are constantly bombarded with stimuli. Poor management of cognitive load can lead to stress, burnout, and decreased quality of life, making it essential for individuals to develop strategies for filtering information and maintaining focus (Sweller et al., 2011). As AI continues to proliferate in everyday life, its role in influencing these psychological constructs cannot be overlooked.

In the context of AI integration, motivation and cognitive load are not merely academic concerns; they are central to how individuals experience and interact with these technologies. AI has the potential to enhance motivation by providing personalized experiences that cater to individual interests, offering customized challenges that keep engagement high. For example, AI-driven platforms in education and fitness can adjust their content to meet users' unique profiles, sustaining motivation and encouraging goal attainment (Goodfellow et al., 2016). Similarly, in professional settings, AI can alleviate cognitive load by automating repetitive tasks, freeing individuals to focus on creative and strategic activities that align with their intrinsic motivations. However, these benefits are not guaranteed. If AI systems are poorly implemented, they may exacerbate cognitive overload or inadvertently erode intrinsic motivation by removing the autonomy that drives personal satisfaction.

The rapid integration of AI raises important questions about how these technologies impact our ability to manage cognitive demands and maintain motivation. For individuals like Alex, who operate within AI-mediated environments, the blend of assistance and control AI offers acts as a mirror. AI helps Alex streamline work processes, yet it also brings moments of introspection—wondering if reliance on these tools is eroding their sense of mastery over the work itself. In this way, AI both facilitates and complicates the human experience, acting as both a resource and a reflection of Alex's evolving identity and capabilities. Chapter 2 will delve deeper into the theoretical foundations underpinning these constructs, examining how motivation and cognitive load influence and are influenced by our interactions with AI. This exploration will provide a foundation for the thematic literature review in Chapter 3, where I will systematically analyze empirical studies on AI's impact on motivation, cognitive load, and identity.

By grounding this inquiry in established psychological theories, the broader implications of AI on human cognition can be more easily understood, along with the continual identification of gaps in understanding. This inquiry will not only highlight the potential benefits of AI but also illuminate the risks of over-reliance on these technologies, particularly as they relate to cognitive and motivational outcomes. Through a careful examination of motivation, cognitive load, and identity, we can gain insight into how AI shapes our understanding of ourselves and our place in an increasingly interconnected world. This exploration will ultimately guide the subsequent analysis of how AI interacts with human motivation, cognitive processing, and identity formation across educational, professional, and personal contexts.

CHAPTER TWO: THE PARADOX OF AI & HUMAN COGNITION

One afternoon, Alex is reviewing a project report generated with AI assistance when they pause, struck by a thought: how much of their work—and even their thought processes—has gradually been shaped by the AI tools they use every day? As they consider the algorithms behind these tools, questions arise about the relationship between human cognition and AI. Is the AI simply supporting their tasks, or is it influencing how they approach problems and make decisions? This moment of reflection reveals a more profound reality: AI's role extends beyond productivity, subtly intertwining with the way they think, learn, and perceive themselves.

Understanding the complex relationship between artificial intelligence (AI) and human cognition requires exploring several foundational concepts. By examining the definitions and historical developments of AI, machine learning, deep learning, motivation, Cognitive Load Theory (CLT), Identity Theory, and human cognition, we can appreciate their interconnections and collective impact on society and academia. This multifaceted approach not only highlights how these technologies are reshaping societal structures but also underscores the importance of grounding AI development in psychological and cognitive theories to foster ethical and effective human-AI interactions.

Historical Development of AI

Artificial intelligence has evolved remarkably since its conceptual origins. Defined as the capability of machines to perform tasks that typically require human intelligence—such as learning, reasoning, problem-solving, and decision-making (Russell & Norvig, 2021)—AI has transitioned from a theoretical concept to a powerful force shaping contemporary society. Alan Turing's seminal work in 1950 laid the modern groundwork for thinking about machines simulating human reasoning, introducing the notion that computational machines could perform any conceivable mathematical operation. Turing posited that “a machine can be constructed which will simulate the behavior of any other machine” (Turing, 1950), a concept that became central to AI research. However, the theoretical roots of these ideas were planted much earlier by Charles Babbage and his collaborator Ada Lovelace, whose insights about the potential of machines influenced early conceptions of computational power.

The Analytical Engine, conceived by Charles Babbage in the 1830s, is often regarded as the first conceptual model of a general-purpose computer. Unlike Babbage’s earlier Difference Engine, which was limited to specific arithmetic tasks, the Analytical Engine was designed to perform any type of calculation or sequence of operations by following a set of programmed instructions stored on punch cards (Hyman, 1982). Babbage envisioned the machine with components remarkably similar to those in modern computers. He described “mills” to process data, functioning much like a modern Central Processing Unit (CPU), a “store” for holding information (analogous to memory), and input/output devices for managing data and results (Swade, 2000). Although the machine was never fully realized, its design laid the conceptual groundwork for future developments in computing, including those that would inspire Turing’s later work.

Ada Lovelace, who collaborated closely with Babbage, played a pivotal role in expanding the theoretical possibilities of the Analytical Engine. In 1843, while translating an article by Luigi Menabrea on the machine, Lovelace added extensive notes, including her famous "Note G" (Toole, 1992). In this note, she outlined an algorithm for calculating Bernoulli numbers, which is recognized as the first computer algorithm, making her the first computer programmer (Fuegi & Francis, 2003). Beyond this technical contribution, Lovelace’s insights into the machine’s capabilities went further than Babbage’s original vision. She argued that the Analytical Engine could manipulate symbols according to rules, enabling it to perform tasks beyond arithmetic, such as composing music, if properly programmed. This revolutionary idea transformed the perception of the Analytical Engine from a mere number-crunching device into a symbolic processor, a notion that foreshadowed the broader applications Turing would later theorize.

The formal establishment of AI as a distinct field occurred at the Dartmouth Conference in 1956, where researchers gathered to explore the possibilities of creating intelligent machines (McCarthy et al., 1956). Here, foundational figures like John McCarthy, Marvin Minsky, and Herbert Simon envisioned a future where machines could perform tasks requiring human-like intelligence. Although the optimistic predictions of early AI researchers remain unfulfilled– similar to the unfulfilled promises of early experimental designs in education– these ambitions laid the groundwork for subsequent advancements (Campbell & Stanley, 1963).

Despite initial enthusiasm, the field has repeatedly experienced periods of stagnation known as AI winters–phases where funding and interest waned due to unmet expectations and disappointing progress. During these AI winters, the excitement surrounding artificial intelligence often cooled, as skepticism grew about the feasibility and practicality of achieving human-like intelligence. However, each winter was eventually followed by an AI summer, when renewed interest, technological breakthroughs, and increased funding brought the field back to life.

The first major AI winter occurred in the late 1970s and early 1980s, as funding for AI research dried up due to unmet expectations. During this period, researchers began to question whether AI could ever achieve human-level intelligence. This skepticism was compounded by technological constraints; computers of that era lacked the processing power necessary for complex AI tasks (Crevier, 1993). As funding waned, AI research shifted toward more practical applications, setting the stage for the resurgence that would come with machine learning.

The 1990s and early 2000s saw a renewed interest in AI, fueled by advances in computing power and the emergence of the internet, which provided vast amounts of data for training AI systems. A pivotal moment came in 1997 when IBM's Deep Blue, a highly advanced chess-playing computer, defeated world chess champion Garry Kasparov. Deep Blue was a specialized system, built specifically to handle the complexities of chess by processing an extraordinary number of possible moves and outcomes per second (Campbell et al., 2002). This achievement was significant because chess had long been considered a game that required uniquely human skills like strategic thinking, intuition, and foresight. Deep Blue's victory demonstrated that AI could outperform human experts in tasks previously thought to require human intelligence, marking a turning point in public perception and the ambitions of AI research. However, while Deep Blue showcased AI's potential, it was still fundamentally limited to a narrow, rule-bound domain that was created by humans. Its approach relied heavily on brute-force computation and domain-specific programming, lacking the flexibility to adapt beyond chess or handle other tasks autonomously. The broader potential of AI would not be fully realized until the development of machine learning algorithms capable of generalizing across diverse data types.

The technological advances found in the 1990s and 2000s would propel this vision into reality. While machine learning began development in the 1980s, the technology boom surrounding Y2K would provide the technological processing required to make machine learning algorithms possible. Instead of being explicitly programmed for a task, machine learning algorithms identify patterns in data and improve their performance over time (Mitchell, 1997).

This approach allows systems to adapt to new situations without reprogramming, making them more flexible and scalable. The advent of big data– massive and complex datasets generated from a variety of sources such as the internet, sensors, and social media– and improvements in computing power further fueled machine learning's growth by providing the vast expanse of data required to train these models effectively. These resources enabled the development of more sophisticated algorithms which could then be applied to real-world scenarios with greater accuracy and versatility.

Machine learning introduced a paradigm shift in AI research, as it enabled systems to learn from data rather than rely on pre-programmed rules. Supervised Learning, a popular form of machine learning, uses labeled data to train algorithms on specific tasks, such as image classification or sentiment analysis. This method has been particularly effective in applications like facial recognition and language translation. In comparison, unsupervised learning deals with unlabeled data and is used to uncover hidden patterns, as seen in clustering algorithms that help segment customers for targeted marketing (Goodfellow et al., 2016). These different approaches demonstrate the versatility of machine learning, which can be tailored to a wide range of applications.

Building upon machine learning, deep learning is a specialized subfield that utilizes artificial neural networks with multiple layers—known as deep neural networks—to model and interpret complex patterns within data (LeCun, Bengio, & Hinton, 2015). At its core, an artificial neural network is a computational model inspired by the human brain's network of neurons. These networks consist of interconnected nodes, or "neurons," organized into layers. Each neuron processes input data and passes information to other neurons, allowing the network to learn and adapt by adjusting the connections, or weights, between neurons.